Analyzing Amazon Employee Sentiments Using Glassdoor Reviews

Introduction

Amazon's employee turnover rate is approximately 150%, which costs the company an estimated $8 billion annually. In order to investigate the root causes of high turnover rate, employee sentiment analysis was conducted using reviews of current and former employees on Glassdoor, a platform which allows employees to leave anonymous reviews.

Data Collection

140 Glassdoor reviews were collected, spanning from 03/16/2023 to 06/03/2025. Data collected included factors such as employee status (current or former), the text of the review, and the review scores. The raw dataset can be found here.

Data Cleaning

The dataset was imported into a Pandas dataframe using a Google Colab notebook. The columns "company_name," "review_advice," and "career_opportunities_rating" were dropped from the dataframe, as company name was the same for every observation, and the other two fields were often left blank. Any rows which had null values in the "summary" field were then dropped as well. Lastly, any duplicates were dropped, and the dataframe was saved as "glassdoor_cleaned.csv."

Next, using the following code, NLTK (Natural Language Toolkit) was used to define a function "clean_text" to remove filler words, convert all text to lowercase, and remove punctuation. This function was then applied to the columns "review_pros," "review_cons," "advice_to_management," and "summary."

```
import nltk
import string
from nltk.corpus import stopwords
nltk.download('stopwords')

def clean_text(text):
    if pd.isnull(text):
        return ""
    tokens = text.lower().translate(str.maketrans('', '', string.punctuation)).split()
    return ' '.join([word for word in tokens if word not in stopwords.words('english')])
```

The resulting columns were named "review_pros_cleaned," "review_cons_cleaned," "advice_to_management_cleaned," and "summary_cleaned" respectively. However, "advice_to_management_cleaned" was then dropped from the dataset as it had too

many null values. The resulting dataframe was then saved as "glassdoor_cleaned_text.csv."

Data Analysis

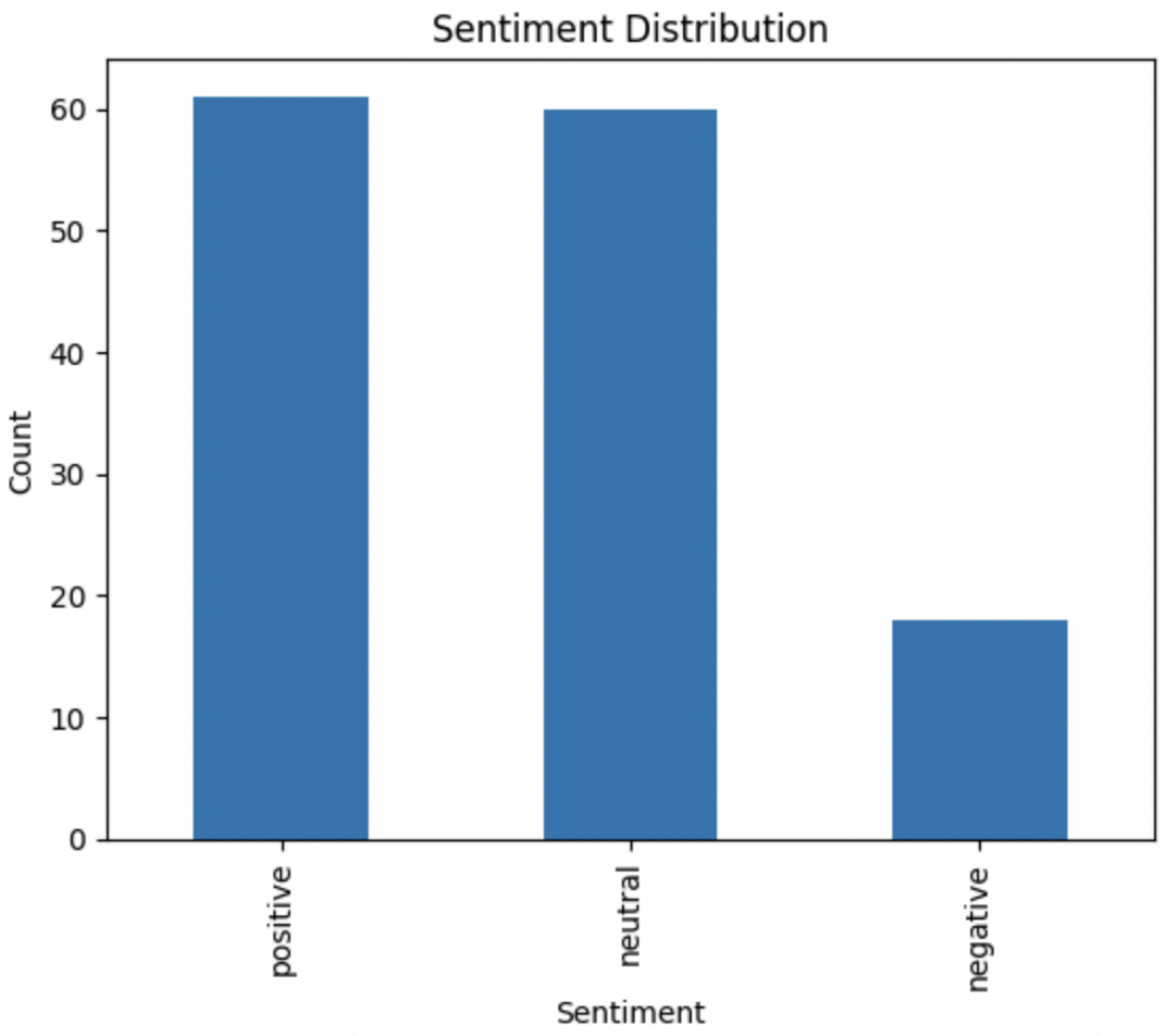
Next, TextBlob was used to analyze the cleaned Glassdoor review data. Using the following code, the "summary_cleaned" data was analyzed using TextBlob to create new columns for sentiment polarity and sentiment subjectivity, named "sentiment_polarity" and "sentiment_subjectivity" respectively. A function "label_sentiment" was then defined to define sentiment as positive if sentiment scores were greater than 0.1, negative if less than -0.1, or otherwise neutral. This function was applied to the "sentiment_polarity" column to generate another new column of sentiment labels, named "sentiment_label."

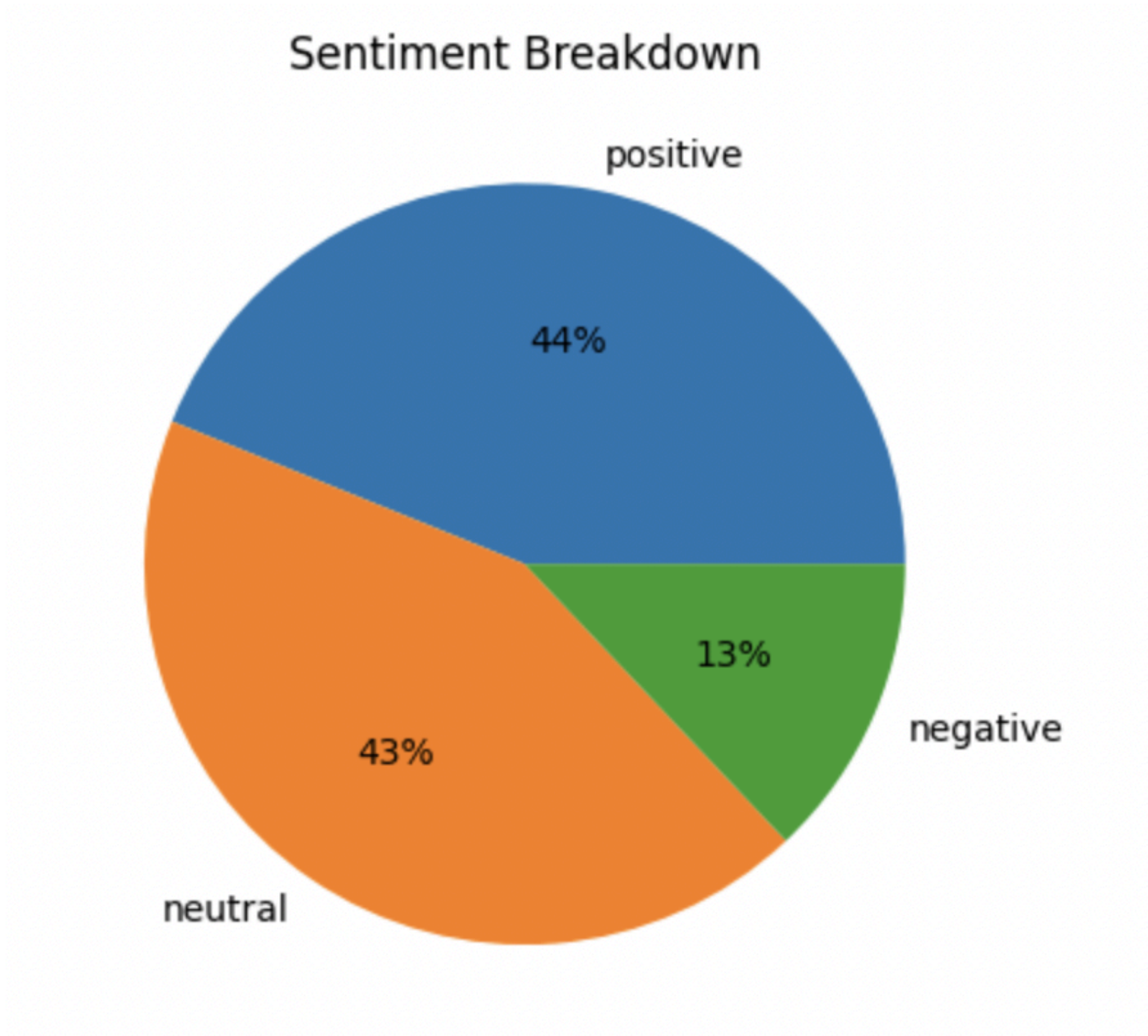
```
df['sentiment_polarity'] = df['summary_cleaned'].apply(lambda x: TextBlob(x).sentiment.polarity)
df['sentiment_subjectivity'] = df['summary_cleaned'].apply(lambda x: TextBlob(x).sentiment.subjectivity)
```

```
def label_sentiment(score):
    if score > 0.1:
        return 'positive'
    elif score < -0.1:
        return 'negative'
    else:
        return 'neutral'

df['sentiment_label'] = df['sentiment_polarity'].apply(label_sentiment)
```

Data Visualization

Sentiment distribution was then plotted using both a bar graph and pie chart, resulting in the following graphs:


Sentiment Distribution
Count
0
10
20
30
40
50
60
positive
neutral
negative
Sentiment



Using the following code, the “summary_cleaned” column was then analyzed for the most frequent unigrams, bigrams, and trigrams:

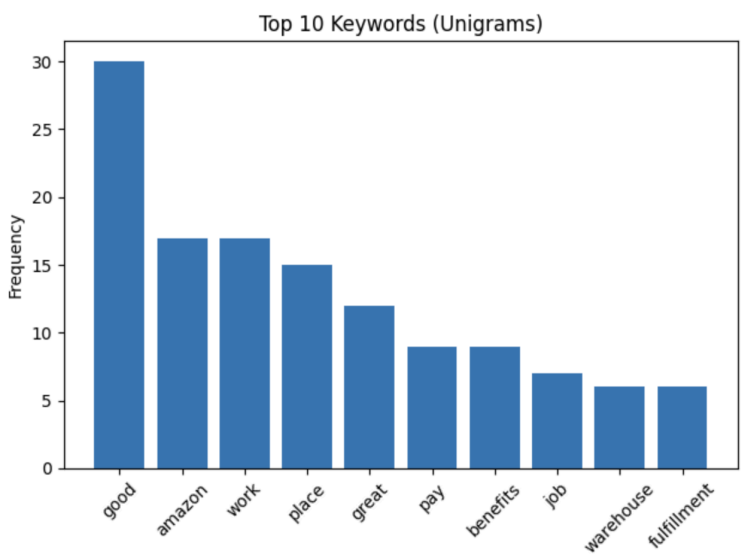
```
from collections import Counter

# Join all cleaned reviews into one big string, then split into words
all_words = ' '.join(df['summary_cleaned']).split()

# Count frequency of each word
word_counts = Counter(all_words)
```

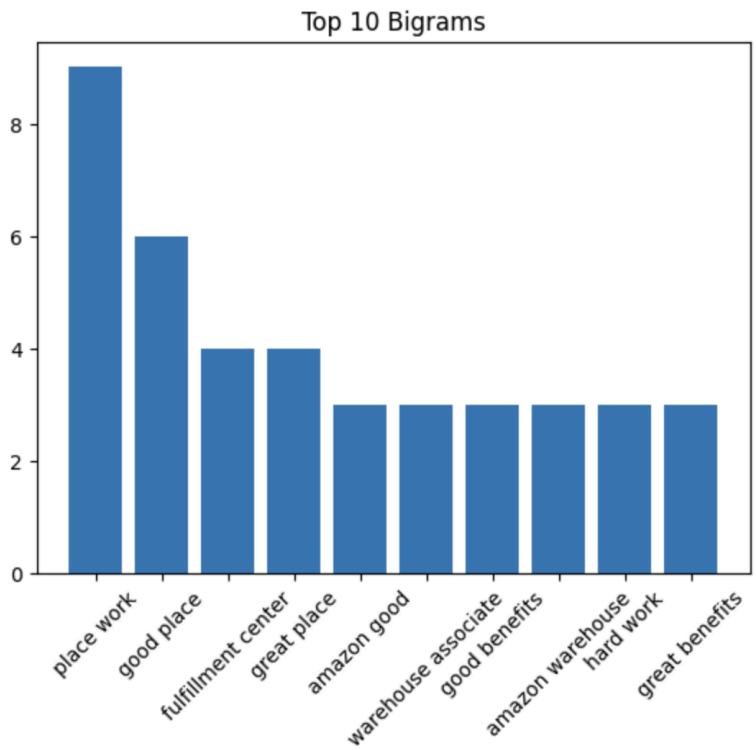
```python
import matplotlib.pyplot as plt

most_common = word_counts.most_common(10)
words, counts = zip(*most_common)

plt.bar(words, counts)
plt.title("Top 10 Keywords (Unigrams)")
plt.xticks(rotation=45)
plt.ylabel("Frequency")
plt.tight_layout()
plt.show()
```

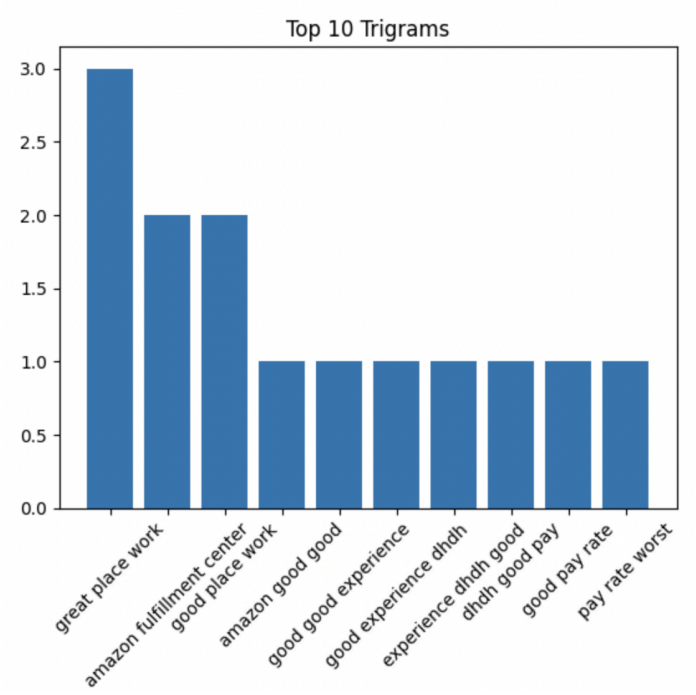
```python
from nltk.util import ngrams

bigrams = list(ngrams(all_words, 2))
trigrams = list(ngrams(all_words, 3))

bigram_counts = Counter(bigrams)
trigram_counts = Counter(trigrams)
```

```python
common_bigrams = bigram_counts.most_common(10)
bg_labels = [' '.join(bg) for bg, count in common_bigrams]
bg_counts = [count for bg, count in common_bigrams]

plt.bar(bg_labels, bg_counts)
plt.title("Top 10 Bigrams")
plt.xticks(rotation=45)
plt.show()
```

```python
common_trigrams = trigram_counts.most_common(10)
tg_labels = [' '.join(bg) for bg, count in common_trigrams]
tg_counts = [count for bg, count in common_trigrams]

plt.bar(tg_labels, tg_counts)
plt.title("Top 10 Trigrams")
plt.xticks(rotation=45)
plt.show()
```







Finally, using the following code a word cloud was generated to visualize the most frequently used words in the “summary_cleaned” column.

```python
from wordcloud import WordCloud

wordcloud = WordCloud(width=800, height=400, background_color='white').generate(' '.join(all_words))

plt.figure(figsize=(15, 7))
plt.imshow(wordcloud, interpolation='bilinear')
plt.axis('off')
plt.title("Keyword Word Cloud")
plt.show()
```

Keyword Word Cloud
good
amazon
great
benefits
place
work
review
place
working
warehouse
job
bad
okay
work
fulfillment
center
associate
salary
company
difficult
experience
opportunity
amazing
terrible
worst
monotonous
hard
decent
easy
money
keep
culture
worker
best
year
okay
trabajo
boring
rate
ok
safety
proper
promote
partner
pretty
less
learn
perform
cool
buena
alright
worth
fc
results
centre
put
trying
procons
von
workplace
methodologies
efficiency
satisfied
comfortable
training
hours
program
monate
fulfillment
time
receiving
hardwork
hr
expérience
técnico
going
muy
cash
cold
sweat
mco1
technical
hamilton
seasonally
excellence
standards
general
thomas
opción
favoritism
bdl2
attitude
mtn1
nothign
meh
oit
fun
lean
maintain
cons
inbound
camp
environment
senior
starters
enviornment
amp
man
intern
high
long
deal
employment
expirience
cost
stow
american
targets
befristet
behandlung
awesome
dont
employee
based
part
first
pros
start
toxicity
unkn
former
super
manager
wasn
sign
heavy
quality
bit
deteriorate
ideal
cow
base
learning
wilmington
yes
rating
operational
dissapointing
rme
t
one
demanding
excellent
nice
buen
mitarbeitern
loved
concentration
lifting
shifts
shop
picker
management
weather
für
sueldo
fastpaced
driving
full
sometimes
physically
income
dhdh
rapidly
greatest
pressure
something
st
cold
cash
muy
okay
técnico
going
year
mco1
best
sweat
worker
technical
hamilton
seasonally
excellence
standards
general
thomas
opción
favoritism
bdl2
attitude
mtn1
nothign
meh
oit
fun
lean
worth
fc
summary
cool
buena
alright
de
safety
proper
promote
partner
pass
learn
perform
pretty
less
amzn
rate
beginner
inbound
camp
environment
maintain
cons
senior
starters
enviornment
amp
man
intern
high
long
deal
employment
expirience
cost
stow
trabajo
american
targets
befristet
behandlung
awesome
dont
company
difficult
employee
based
part
first
pros
start
toxicity
unkn
former
super
manager
wasn
sign
heavy
quality
bit
deteriorate
ideal
cow
base
learning
wilmington
yes
rating
operational
dissapointing
rme
t
one
demanding
excellent
nice
buen
mitarbeitern
loved
concentration
lifting
shifts
easy
money
keep
program
monate
fulfillment
time
receiving
hardwork
hr
expérience
técnico
going
muy
cash
cold
sweat
mco1